EE4-18
EE9-AO6

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE PART IV: MEng and ACGI

Corrected copy

RADIO FREQUENCY ELECTRONICS

Monday, 13 May 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Lucyszyn

Second Marker(s) : W.T. Pike

© Imperial College London

Special instructions for invigilators

This is a closed book examination.

Special instructions for students

All variable have their usual meaning.

Standard filter tables are given at the back.

© Imperial College London

The Questions

1.

a) Consider the simplified equivalent lumped-element model of a MESFET.

i) Draw the equivalent circuit model that only has $R_g = 2\ \Omega, C_{gs} = 50$ pF, $g_m = 100$ mS and $R_{ds} = 50$ kΩ. What can you infer about the signal levels and useful frequency range of this model?

[2]

ii) What is the reverse voltage-wave transmission coefficient for the model in 1(a)(i)?

[1]

iii) For the model in 1(a)(i), connected to an input generator consisting of a voltage source V_S and source impedance $R_S = 50\ \Omega$, derive the input transfer function V_{gs}/V_{IN} and describe the corresponding equivalent circuit model in terms of a filter.

[2]

iv) From the model in 1(a)(iii), calculate the input time constant and relaxation frequency?

[2]

v) With log-log scales, sketch the magnitude of the transfer function obtained in 1(a)(iii) in terms of power response against normalized frequency between 0.1 and 100. Overlay the corresponding Bode Plot and quantify the frequency behaviour.

[3]

vi) From the transfer function obtained in 1(a)(iii), derive an expression for the differential-phase group delay. With log-log scales, sketch this function against normalized frequency between 0.1 and 100.

[3]

b) For the MESFET model in 1(a)(i), connected to the 50 Ω source impedance, design a suitable input lumped-element matching circuit for operation at 1 MHz and calculate the theoretical bandwidth for this amplifier.

[4]

c) For the MESFET model in 1(a)(i), briefly explain the concepts of AM-to-PM conversion and AM-to-AM conversion. Illustrate the resulting error vector magnitude when both types of conversion effect symbols in an I-Q constellation with a digitally modulated signal.

[3]

2.

a) Consider the satellite communications link illustrated in Figure 2.1. The satellite is in a circular geosynchronous orbit in the plane of the Earth's equator and has an altitude of 35,786 km above see level. Both the uplink and downlink earth stations are both located on the equator, at sea level. For simplicity, ignore the curvature of the earth.

i) Write down the well-known Friis' range equation for a point-to-point communications link.

[2]

ii) From 2(a)(i), deduce the expression for the Space-loss term shown in Figure 2.1 and use this to calculate the uplink and downlink ranges.

[2]

iii) What is the separation distance between the uplink and downlink earth stations?

[1]

iv) If the uplink and downlink earth station antennas are now pointed directly towards each other and the downlink earth station is retuned to operate at 6 GHz, calculate its LNA gain to achieve the specified output signal power.

[2]

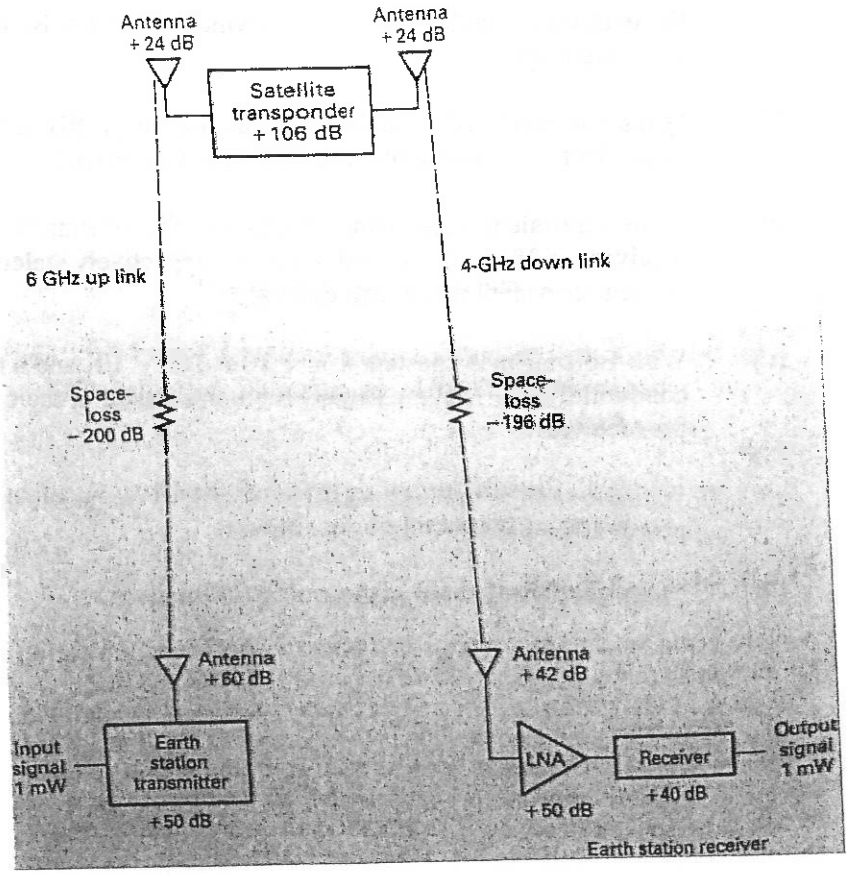


Figure 2.1 (taken from R.L. Douglas, *Satellite Communications Technology*).

Question 2 continues onto the next page...

b) Consider a signal generator connected to a load impedance, via a transmission line having a power loss L_L, all being matched to the same reference impedance. Assuming the signal generator, transmission line and load impedance are all in thermodynamic equilibrium and at the physical ambient temperature T_o. The input signal power S_i and input noise power N_i to the transmission line can be combined to give an input signal-to-noise power ratio $SNR = S_i/N_i$.

i) Give the mathematical relationship between equivalent noise temperature and noise factor.

[1]

ii) Give an expression for the output noise power.

[1]

iii) If the transmission line has a loss of -3 dB, what is its equivalent noise temperature in relation to T_o?

[1]

iv) Determine the output signal power S_o and output noise power N_o from the transmission line and give the output signal-to-noise power ratio $SNR = S_o/N_o$. What is the mathematical relationship between the output and input SNR?

[2]

c) Consider the downlink earth station in Figure 2.1.

i) Write down the well-known Friss' formula for the noise factor of cascaded stages.

[1]

ii) Using the relationship in 2(b)(i) with that in 2(c)(i), derive the equivalent noise temperature for two cascaded stages.

[2]

iii) If the equivalent noise temperatures for the antenna, LNA and receiver are 38 K, 120 K and 4,000 K, respectively, calculate the system equivalent noise temperature.

[2]

iv) With Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K and a receiver bandwidth of 27 MHz, calculate the noise power at the output of the receiver.

[1]

v) With the satellite having an EIRP of +34 dBW, calculate the signal power at the output of the receiver.

[1]

vi) Calculate the SNR at the output of the receiver.

[1]

3.

a) Explain the difference between half-duplex and full-duplex transceiver architectures in how they achieve Tx-Rx isolation and give the name of an international standard for each used in modern mobile communications. Briefly describe the difference between a duplexer, diplexer and triplexer.

[4]

b) Draw the basic block diagram of a Direct Conversion receiver, labelling each block and describe its attributes and drawbacks.

[10]

c) In terms of sidebands, explain when quadrature conversion is required. Give an example of one analogue and one digital modulation scheme that does not require quadrature conversion and one analogue and one digital modulation scheme that requires quadrature conversion.

[2]

d) Draw the basic block diagram of a Direct Conversion transmitter with I-Q digital modulator.

[4]

© Imperial College London

4.

a) Draw the conventional symbol for a circulator, indicating each port number, and derive the corresponding S-parameter matrix for an ideal isolator. Clearly explain any assumptions.

[6]

b) Given a non-ideal circulator, draw a single-stage reflection topology. By inspection, derive expressions for the corresponding S-parameter matrix. The following general mathematical relationship can be used to represent infinite reflections between two impedance boundaries:

$$\sum_{i=0}^{\infty} A^i = \frac{1}{1-A} \tag{4.1}$$

[5]

c) Simplify the S-parameter matrix derived in 4(b) by assuming the circulator has 3-way symmetry.

[3]

d) Simplify the S-parameter matrix derived in 4(c) by assuming the circulator is ideal.

[2]

e) How can you implement a variable phase shifter and a variable attenuator, using monolithic technology and hybrid solutions?

[4]

5.

a) Design a 4th-order Chebychev band pass filter with 0.5 dB passband ripple, having a centre frequency of 2 GHz and -3 dB bandwidth of 100 MHz, for insertion between 50 Ω source and 99.2 Ω load impedances. What is the theoretical worst-case return loss at centre frequency for this filter?

[10]

b) Figure 5.1 shows a 2 GHz 'all-transistor' MMIC 3rd-order Chebyshev band pass filter employing three active inductors. List five advantages and five disadvantages for the practical implementation of this filter implementation.

[10]

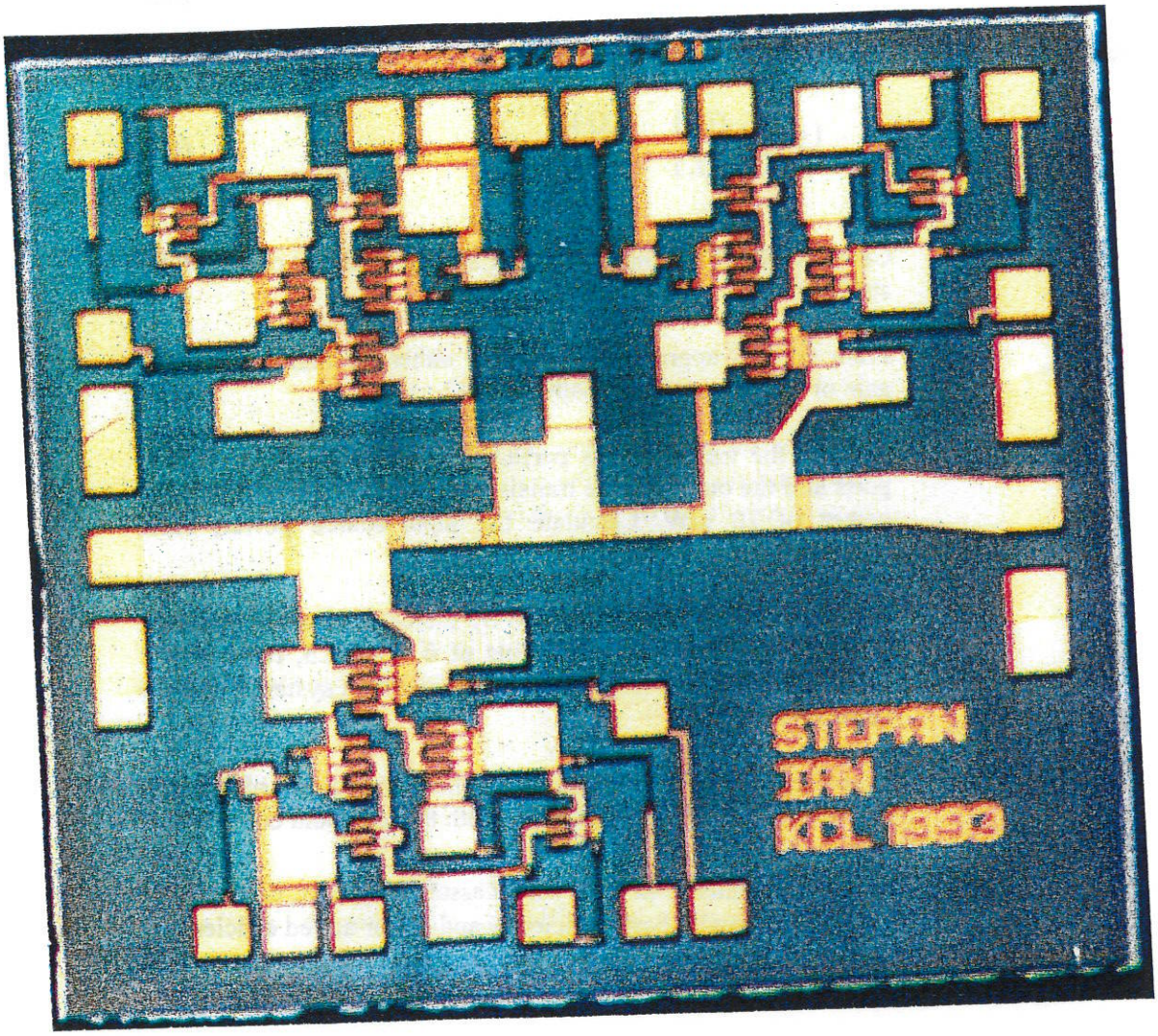


Figure 5.1 2 GHz 'all-transistor' MMIC 3rd-order Chebyshev band pass filter.

© Imperial College London

6. The Rollet stability factor K and unilateral transducer gain G_{TU} for a 2-port network are given by:

$$K = \frac{1-|S_{11}|^2-|S_{22}|^2+|\Delta|^2}{2|S_{12}S_{21}|} \qquad \Delta = S_{11}S_{22} - S_{12}S_{21} \tag{6.1}$$

$$G_{TU} = \frac{|S_{21}|^2(1-|\Gamma_S|^2)(1-|\Gamma_L|^2)}{|(1-S_{11}\Gamma_S)(1-S_{22}\Gamma_L)|^2} \tag{6.2}$$

The network is a transistor having S-parameter matrix [S]:

$$[S] = \begin{bmatrix} 0.1 & 10 \\ 0.1 & 0.1 \end{bmatrix} \tag{6.3}$$

i) Calculate the Rollet stability factor and establish the condition for stability.

[3]

ii) From 6(i), as appropriate, calculate either the maximum stable gain or the maximum available gain.

[3]

iii) Assume the transistor has perfect impedance matching at both ports and the output of the transistor is 10 mW with an input DC power of 20 mW. Calculate the input source power, basic efficiency and power-added efficiency.

[3]

iv) This transistor is now terminated by source and load impedances that exhibit reflection coefficients of 0.1 and 0.5, respectively. Calculate the source and load impedances, for a system reference impedance of 50 Ω.

[4]

v) With the scenario in 6(iv), calculate the unilateral transducer gain and individual contributions from both the input and out gains.

[4]

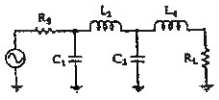
vi) Using the results from 6(v), which assumes $S_{12} = 0$, calculate the new output power, basic efficiency and power-added efficiency.

[3]

© Imperial College London

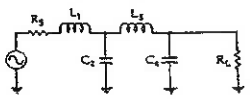
Standard Filter Tables

Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple

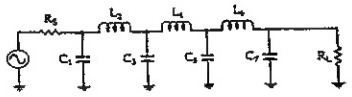


n	R_s/R_L	C_1	L_2	C_3	L_4
2	1.984	0.983	1.950		
	2.000	0.909	2.103		
	2.500	0.584	3.185		
	3.333	0.375	4.411		
	5.000	0.228	6.700		
	10.000	0.105	13.322		
	∞	1.307	0.975		
3	1.000	1.864	1.280	1.834	
	0.900	1.918	1.309	2.028	
	0.800	1.997	1.120	2.237	
	0.700	2.114	1.015	2.517	
	0.500	2.557	0.759	3.436	
	0.400	2.985	0.615	4.242	
	0.300	3.729	0.463	5.576	
	0.200	5.254	0.309	8.225	
	0.100	9.890	0.153	16.118	
	∞	1.574	1.518	0.932	
4	1.984	0.920	2.586	1.304	1.828
	2.000	0.845	2.720	1.238	1.985
	2.500	0.516	3.766	0.869	3.121
	3.333	0.344	5.120	0.621	4.480
	5.000	0.210	7.708	0.400	6.987
	10.000	0.098	15.352	0.194	14.262
	∞	1.436	1.889	1.521	0.913

n	R_L/R_s	L_1	C_2	L_3	C_4

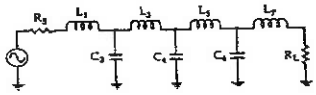


Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple



n	R_s/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	1.807	1.303	2.891	1.303	1.807		
	0.900	1.854	1.222	2.849	1.238	1.970		
	0.800	1.926	1.126	3.060	1.157	2.185		
	0.700	2.035	1.015	3.353	1.058	2.470		
	0.600	2.200	0.890	3.785	0.942	2.861		
	0.500	2.457	0.754	4.387	0.810	3.414		
	0.400	2.870	0.609	5.296	0.664	4.245		
	0.300	3.588	0.459	6.871	0.508	5.625		
	0.200	5.064	0.306	10.054	0.343	8.367		
	0.100	9.556	0.153	19.847	0.173	16.574		
	∞	1.630	1.740	1.922	1.514	0.903		
6	1.984	0.905	2.577	1.368	2.713	1.299	1.796	
	2.000	0.830	2.704	1.291	2.872	1.237	1.956	
	2.500	0.506	3.722	0.890	4.109	0.881	3.103	
	3.333	0.337	5.055	0.632	5.899	0.635	4.481	
	5.000	0.206	7.615	0.406	8.732	0.412	7.031	
	10.000	0.096	15.186	0.197	17.681	0.203	14.433	
7	1.000	1.790	1.296	2.718	1.385	2.718	1.296	1.790
	0.900	1.835	1.215	2.869	1.308	2.883	1.234	1.953
	0.800	1.905	1.118	3.076	1.215	3.107	1.153	2.188
	0.700	2.011	1.007	3.384	1.105	3.416	1.058	2.455
	0.600	2.174	0.882	3.772	0.979	3.852	0.944	2.848
	0.500	2.428	0.747	4.370	0.838	2.289	0.814	3.405
	0.400	2.835	0.604	5.295	0.685	5.470	0.669	4.243
	0.300	3.548	0.455	6.887	0.522	7.134	0.513	5.635
	0.200	5.007	0.303	10.049	0.352	10.496	0.348	8.404
	0.100	9.458	0.151	19.849	0.178	20.631	0.176	16.665
	∞	1.646	1.777	2.031	1.789	1.924	1.503	0.895

n	R_L/R_s	L_1	C_2	L_3	C_4	L_5	C_6	L_7



© Imperial College London